

Quantitative Discovery of Qualitative Information: A General Purpose Document Clustering Methodology

Gary King
Institute for Quantitative Social Science
Harvard University

joint work with
Justin Grimmer (Harvard)

(talk at IQSS Mindich Text Analysis Conference, 5/30/09)

The Problem: Discovery from Unstructured Text

- Examples: scholarly literature, news stories, medical information, blog posts, comments, product reviews, emails, social media updates, audio-to-text summaries, speeches, press releases, legal decisions, etc.

The Problem: Discovery from Unstructured Text

- Examples: scholarly literature, news stories, medical information, blog posts, comments, product reviews, emails, social media updates, audio-to-text summaries, speeches, press releases, legal decisions, etc.
- 10 minutes of worldwide email = 1 LOC equivalent

The Problem: Discovery from Unstructured Text

- Examples: scholarly literature, news stories, medical information, blog posts, comments, product reviews, emails, social media updates, audio-to-text summaries, speeches, press releases, legal decisions, etc.
- 10 minutes of worldwide email = 1 LOC equivalent
- An essential part of discovery is **classification**: “one of the most central and generic of all our conceptual exercises. . . the foundation not only for conceptualization, language, and speech, but also for mathematics, statistics, and data analysis. . . Without classification, there could be no advanced conceptualization, reasoning, language, data analysis or, for that matter, social science research.” (Bailey, 1994).

The Problem: Discovery from Unstructured Text

- Examples: scholarly literature, news stories, medical information, blog posts, comments, product reviews, emails, social media updates, audio-to-text summaries, speeches, press releases, legal decisions, etc.
- 10 minutes of worldwide email = 1 LOC equivalent
- An essential part of discovery is **classification**: “one of the most central and generic of all our conceptual exercises. . . the foundation not only for conceptualization, language, and speech, but also for mathematics, statistics, and data analysis. . . Without classification, there could be no advanced conceptualization, reasoning, language, data analysis or, for that matter, social science research.” (Bailey, 1994).
- We focus on **cluster analysis**: discovery through (1) classification and (2) simultaneously inventing a classification scheme

The Problem: Discovery from Unstructured Text

- Examples: scholarly literature, news stories, medical information, blog posts, comments, product reviews, emails, social media updates, audio-to-text summaries, speeches, press releases, legal decisions, etc.
- 10 minutes of worldwide email = 1 LOC equivalent
- An essential part of discovery is **classification**: “one of the most central and generic of all our conceptual exercises. . . the foundation not only for conceptualization, language, and speech, but also for mathematics, statistics, and data analysis. . . Without classification, there could be no advanced conceptualization, reasoning, language, data analysis or, for that matter, social science research.” (Bailey, 1994).
- We focus on **cluster analysis**: discovery through (1) classification and (2) simultaneously inventing a classification scheme
- (We analyze text; our methods apply more generally)

Why Johnny Can't Classify (Optimally)

- $\text{Bell}(n)$ = number of ways of partitioning n objects

Why Johnny Can't Classify (Optimally)

- $\text{Bell}(n)$ = number of ways of partitioning n objects
- $\text{Bell}(2) = 2$ (AB, A B)

Why Johnny Can't Classify (Optimally)

- $\text{Bell}(n)$ = number of ways of partitioning n objects
- $\text{Bell}(2) = 2$ (AB, A B)
- $\text{Bell}(3) = 5$ (ABC, AB C, A BC, AC B, A B C)

Why Johnny Can't Classify (Optimally)

- $\text{Bell}(n)$ = number of ways of partitioning n objects
- $\text{Bell}(2) = 2$ (AB, A B)
- $\text{Bell}(3) = 5$ (ABC, AB C, A BC, AC B, A B C)
- $\text{Bell}(5) = 52$

Why Johnny Can't Classify (Optimally)

- $\text{Bell}(n)$ = number of ways of partitioning n objects
- $\text{Bell}(2) = 2$ (AB, A B)
- $\text{Bell}(3) = 5$ (ABC, AB C, A BC, AC B, A B C)
- $\text{Bell}(5) = 52$
- $\text{Bell}(100) \approx$

Why Johnny Can't Classify (Optimally)

- $\text{Bell}(n)$ = number of ways of partitioning n objects
- $\text{Bell}(2) = 2$ (AB, A B)
- $\text{Bell}(3) = 5$ (ABC, AB C, A BC, AC B, A B C)
- $\text{Bell}(5) = 52$
- $\text{Bell}(100) \approx 10^{28} \times \text{Number of elementary particles in the universe}$

Why Johnny Can't Classify (Optimally)

- $\text{Bell}(n)$ = number of ways of partitioning n objects
- $\text{Bell}(2) = 2$ (AB, A B)
- $\text{Bell}(3) = 5$ (ABC, AB C, A BC, AC B, A B C)
- $\text{Bell}(5) = 52$
- $\text{Bell}(100) \approx 10^{28} \times$ Number of elementary particles in the universe
- Now imagine choosing the *optimal* classification scheme by hand

Why Johnny Can't Classify (Optimally)

- $\text{Bell}(n)$ = number of ways of partitioning n objects
- $\text{Bell}(2) = 2$ (AB, A B)
- $\text{Bell}(3) = 5$ (ABC, AB C, A BC, AC B, A B C)
- $\text{Bell}(5) = 52$
- $\text{Bell}(100) \approx 10^{28} \times$ Number of elementary particles in the universe
- Now imagine choosing the *optimal* classification scheme by hand
- Qualitative-only approaches are hopeless

Why Johnny Can't Classify (Optimally)

- $\text{Bell}(n)$ = number of ways of partitioning n objects
- $\text{Bell}(2) = 2$ (AB, A B)
- $\text{Bell}(3) = 5$ (ABC, AB C, A BC, AC B, A B C)
- $\text{Bell}(5) = 52$
- $\text{Bell}(100) \approx 10^{28} \times \text{Number of elementary particles in the universe}$
- Now imagine choosing the *optimal* classification scheme by hand
- Qualitative-only approaches are hopeless
- That we think of all this as astonishing . . . is astonishing

Why HAL Can't Classify Either

Why HAL Can't Classify Either

- The Goal — an optimal application-independent cluster analysis method — is mathematically impossible:

Why HAL Can't Classify Either

- The Goal — an optimal application-independent cluster analysis method — is mathematically impossible:
 - **No free lunch theorem**: every possible clustering method performs equally well on average over all possible substantive applications

Why HAL Can't Classify Either

- The Goal — an optimal application-independent cluster analysis method — is mathematically impossible:
 - **No free lunch theorem**: every possible clustering method performs equally well on average over all possible substantive applications
- Existing methods:

Why HAL Can't Classify Either

- The Goal — an optimal application-independent cluster analysis method — is mathematically impossible:
 - **No free lunch theorem**: every possible clustering method performs equally well on average over all possible substantive applications
- Existing methods:
 - **Many choices**: model-based, subspace, spectral, grid-based, graph-based, fuzzy k -modes, affinity propagation, self-organizing maps,...

Why HAL Can't Classify Either

- The Goal — an optimal application-independent cluster analysis method — is mathematically impossible:
 - **No free lunch theorem**: every possible clustering method performs equally well on average over all possible substantive applications
- Existing methods:
 - **Many choices**: model-based, subspace, spectral, grid-based, graph-based, fuzzy k -modes, affinity propagation, self-organizing maps, . . .
 - **Well-defined** statistical, data analytic, or machine learning foundations

Why HAL Can't Classify Either

- The Goal — an optimal application-independent cluster analysis method — is mathematically impossible:
 - **No free lunch theorem**: every possible clustering method performs equally well on average over all possible substantive applications
- Existing methods:
 - **Many choices**: model-based, subspace, spectral, grid-based, graph-based, fuzzy k -modes, affinity propagation, self-organizing maps, . . .
 - **Well-defined** statistical, data analytic, or machine learning foundations
 - How to add substantive knowledge:

Why HAL Can't Classify Either

- The Goal — an optimal application-independent cluster analysis method — is mathematically impossible:
 - **No free lunch theorem**: every possible clustering method performs equally well on average over all possible substantive applications
- Existing methods:
 - **Many choices**: model-based, subspace, spectral, grid-based, graph-based, fuzzy k -modes, affinity propagation, self-organizing maps,...
 - **Well-defined** statistical, data analytic, or machine learning foundations
 - How to add substantive knowledge: With few exceptions, **who knows?!**

Why HAL Can't Classify Either

- The Goal — an optimal application-independent cluster analysis method — is mathematically impossible:
 - **No free lunch theorem**: every possible clustering method performs equally well on average over all possible substantive applications
- Existing methods:
 - **Many choices**: model-based, subspace, spectral, grid-based, graph-based, fuzzy k -modes, affinity propagation, self-organizing maps,...
 - **Well-defined** statistical, data analytic, or machine learning foundations
 - How to add substantive knowledge: With few exceptions, **who knows?!**
 - The literature: **little guidance on when methods apply**

Why HAL Can't Classify Either

- The Goal — an optimal application-independent cluster analysis method — is mathematically impossible:
 - **No free lunch theorem**: every possible clustering method performs equally well on average over all possible substantive applications
- Existing methods:
 - **Many choices**: model-based, subspace, spectral, grid-based, graph-based, fuzzy k -modes, affinity propagation, self-organizing maps,...
 - **Well-defined** statistical, data analytic, or machine learning foundations
 - How to add substantive knowledge: With few exceptions, **who knows?!**
 - The literature: **little guidance on when methods apply**
 - **Deriving such guidance**: difficult or impossible

Why HAL Can't Classify Either

- The Goal — an optimal application-independent cluster analysis method — is mathematically impossible:
 - **No free lunch theorem**: every possible clustering method performs equally well on average over all possible substantive applications
- Existing methods:
 - **Many choices**: model-based, subspace, spectral, grid-based, graph-based, fuzzy k -modes, affinity propagation, self-organizing maps, . . .
 - **Well-defined** statistical, data analytic, or machine learning foundations
 - How to add substantive knowledge: With few exceptions, **who knows?!**
 - The literature: **little guidance on when methods apply**
 - **Deriving such guidance: difficult or impossible**
 - (Perhaps true by definition in unsupervised learning: If we knew the DGP, we wouldn't be at the discovery stage.)

If Ex Ante doesn't work, try Ex Post

If Ex Ante doesn't work, try Ex Post

- **Methods and substance must be connected** (no free lunch theorem)

If Ex Ante doesn't work, try Ex Post

- **Methods and substance must be connected** (no free lunch theorem)
- The usual approach fails: can't do it by understanding the model

If Ex Ante doesn't work, try Ex Post

- **Methods and substance must be connected** (no free lunch theorem)
- The usual approach fails: can't do it by understanding the model
- We do it **ex post** (by qualitative choice)

If Ex Ante doesn't work, try Ex Post

- **Methods and substance must be connected** (no free lunch theorem)
- The usual approach fails: can't do it by understanding the model
- We do it **ex post** (by qualitative choice)
 - For discovery (our goal): No problem

If Ex Ante doesn't work, try Ex Post

- **Methods and substance must be connected** (no free lunch theorem)
- The usual approach fails: can't do it by understanding the model
- We do it **ex post** (by qualitative choice)
 - For discovery (our goal): No problem
 - For estimation & confirmation: more difficult or biased

If Ex Ante doesn't work, try Ex Post

- **Methods and substance must be connected** (no free lunch theorem)
- The usual approach fails: can't do it by understanding the model
- We do it **ex post** (by qualitative choice)
 - For discovery (our goal): No problem
 - For estimation & confirmation: more difficult or biased
- Complicated concepts are easier to define ex post:

If Ex Ante doesn't work, try Ex Post

- **Methods and substance must be connected** (no free lunch theorem)
- The usual approach fails: can't do it by understanding the model
- We do it **ex post** (by qualitative choice)
 - For discovery (our goal): No problem
 - For estimation & confirmation: more difficult or biased
- Complicated concepts are easier to define ex post:
 - “I know it when I see it” (Justice Stewart's definition of obscenity)

If Ex Ante doesn't work, try Ex Post

- **Methods and substance must be connected** (no free lunch theorem)
- The usual approach fails: can't do it by understanding the model
- We do it **ex post** (by qualitative choice)
 - For discovery (our goal): No problem
 - For estimation & confirmation: more difficult or biased
- Complicated concepts are easier to define ex post:
 - “I know it when I see it” (Justice Stewart's definition of obscenity)
 - Anchoring Vignettes (on defining concepts by example)

If Ex Ante doesn't work, try Ex Post

- **Methods and substance must be connected** (no free lunch theorem)
- The usual approach fails: can't do it by understanding the model
- We do it **ex post** (by qualitative choice)
 - For discovery (our goal): No problem
 - For estimation & confirmation: more difficult or biased
- Complicated concepts are easier to define ex post:
 - “I know it when I see it” (Justice Stewart's definition of obscenity)
 - Anchoring Vignettes (on defining concepts by example)
- **But how to choose from an enormous list of clusterings?**

Our Idea: Meaning Through Geography

Our Idea: Meaning Through Geography

wide at SuperPages.com

195

Car

C

87 566-1282
81 447-4101
800 257-9981
87 566-1282
87 364-5188
361-0380
87 566-4548
87 628-8248
87 445-5116
87 822-2982
87 627-5712
87 569-2698
87 667-5190
87 549-1417
87 338-9110
87 823-9195
87 296-1593
87 670-2078
87 623-9001
87 296-4725
87 542-1521
87 364-5232
87 541-5649
87 739-2662
87 879-0030
87 541-3948
87 446-1513
87 496-4119
800 569-8782

Carriage New England Inc
36 Allen St Ipswich 01908... 978 356-9960
Cartagena Lydia
19 Jewett Ave 02121... 617 323-7639
Cartagena Arith
8 Everett Ave 02119... 617 442-9780
B Hot 02136... 617 361-5253
Jestica 50 Decatur Cir 02129... 617 241-0152
Lucilla 124 Harvard Cam 02139... 617 491-5621
M 95 Snow Rd 02133... 617 323-9713
Melvin 101 Green Cam 02139... 617 576-1061
Carter Nicholas
18 Appleton Street 02114... 617 695-6996
Cartegena O «Harvard Bus 02118... 617 338-8219
Carter Thos J Sr & Claire
1 Paradise Rd 02186... 617 698-6163
50 Thompson Ln 02186... 617 696-6919
Carter A 02131... 617 327-2257
A Industry... 617 442-5230
A 31 Redwine Wy 02129... 617 442-1219
A 280 Putnam Ave Cambridge 02139... 617 492-4174
A M 255 Marchesa Ave 02135... 617 266-7153
Adams 381 Centre St 02186... 617 698-9074
Alice 108 Elmwood Ave 02135... 617 425-0193
Alice «Market Cambridge 02139... 617 945-2711
Andrew F «Vital Air Serv 02143... 617 625-7623
Carter Anne MD
1118 Beacon Dr 02446... 617 739-1022
Carter Athens
777 Newbury Boston 02114... 617 536-6329
B E «St Graduate Ave Mart 02126... 617 296-6911
Carter Barbara L MD
100 New England Medical Center Bldg 02111
Call... 617 636-0051
Carter Becky R 02114... 617 523-4368
Bernard J
121 Cedar St E 02128... 617 567-3430
Bithiah 25 Midway Dr 02124... 617 298-8713
Blake 16 W Vernon St 02188... 617 367-9931
Carter BroadCastling Co
20 Park Pls Box 02116... 617 423-0210
Carter & Burgess Consultants Inc
71 East C St 02141... 617 225-0200
Carter C 2000 Commonwealth Ave 02135... 617 782-2118
C 229 Park Ave 02114... 617 569-1545
C 359 Harvard Cam 02138... 617 491-4822
C 618 Brook Hts 02126... 617 296-6392
C B M «St Cambridge Ave 02138... 617 526-9536

Carter F 31 Wilcox Ave 02133... 617 327-1105
Faye & Ricky
257 Columbus Ave Box 02116... 617 417-7331
Francis S 184 Temple W 02135... 617 323-6761
Franklin & Anne
221 Mt Auburn Cam 02138... 617 354-6798
Fred «Newford Jwn 02130... 617 524-3078
Fred 50 Winchey Rd 02136... 617 698-1343
G & R «Verard Box 02124... 617 436-9906
G T 27 Franklin Ave 02145... 617 623-7121
Gayle 25 Frontview Dr 02134... 617 825-0322
Geo S 115 West Hill Ave 02138... 617 522-3215
George 125 Nausha Box 02114... 617 367-9548
Carter Halliday Associates
107 S Street Box 02111... 617 456-1689
Carter Harry E
38 Irving St Box W 02138... 617 325-5465
Carter Hide Co Inc
46 Summer Box 02130... 617 542-7987
Carter Hilary «J Harvey Cam 02186... 617 876-2750
Horace
942 Walnut Ave 02129... 617 442-5307
Howard Jr 20 Notre Dame St 02119... 617 445-5552
J Cam... 617 354-2688
J 15 Chatham Box 02446... 617 232-7990
J 518 Harvard Box 02446... 617 730-9483
J 775 Ws Play West 02132... 617 323-5574
Carter J «Jacques MD
1 Brookline Pl Box 02446... 617 735-8787
Carter J M
1435 Columbia St S Box 02127... 617 464-1040
Carter J M «Mental Ironworks
Call... 617 436-5353
Carter J Neal Co
40 Newmarket St Box 02116... 617 442-1775
Carter James
1573 Cambridge St Cam 02138... 617 492-1214
James 182 Fisher Ave 02139... 617 739-2193
James
27 Gold Star Rd Cambridge 02140... 617 876-8841
Jas L 14 Broadway Rd Mart 02138... 617 361-0773
Jane 114 Adams Rd Newton 02460... 617 964-0435
Jeffrey «J Warren Ave Box 02116... 617 425-5999
John 115 Mainfield Rd 02119... 617 987-2163
John 127 Summer Box 02120... 617 423-4334
John «Winfield Rd Box 02125... 617 252-1235
June O 329 A Summit Ave Box 02136... 617 734-6109
K 18 Browning Ave Dorchester 02224... 617 285-9456
L 27 Temple Dorchester 02219... 617 282-1593

Carter Nella E
331 Manchester Ave Box 02115... 617 267-6483
Nicholas S F
115 Hampshire Ave 02186... 617 698-5307
Nick 21 Fairfield Box 02116... 617 267-5222
Nick & Debbie
194 Herrick Rd Newton 02459... 617 527-0480
Nicole
38 Chittenden Dr 02122... 617 822-1203
P «Crestwood Pl Box 02135... 617 427-4754
P E 101 E South St Box 02027... 617 268-4213
P L 44 Hutchings Box 02121... 617 427-0170
P R «St Boyer Box 02138... 617 983-6092
Paul & Constance
114 Aronson Ave W Box 02138... 617 325-2036
Paul E 101 E South St Box 02027... 617 268-4546
Paul M 27 Union Bld 02135... 617 787-2115
Carter Pike Driving Inc 17 Beaver Ct
Franklin 02102... 617 235-8488
Carter Prudence
48 Franklin Waterdown 02127... 617 393-3782
Prudence
49 Franklin Waterdown 02127... 617 926-7063
Reginald
386 Braintree Dorchester 02122... 617 541-2843
Renée & Andrew
33 Walnut Box 02108... 617 720-3765
Carter Rice David
Bulfinch Station Postoffice 313 Main Wilmington 01887
Toll Free-Call T & Then... 800 638-1671
Carter S «Industrial Prod 313 Main Wilmington
Toll Free-Call T & Then... 800 619-7447
Carter Se-Printing 613 Main Wilmington
Toll Free-Call T & Then... 800 648-7447
Headquarters 613 Main Wilmington 01887
Call... 978 968-7447
Insignia Center 163 Main Wilmington 01887
Toll Free-Call T & Then... 800 638-1673
Carter Richard
1819 Commonwealth Ave Brighton 02108... 617 987-0836
Richard A 97 Mt Vernon St Box 02038... 617 566-7593
Carter Richard A MD
176 Commonwealth Ave Box 02116... 617 267-0710
Carter Richard K
15 Worcester St Box 02137... 617 268-0448
Robert L 175 Schinola Ave Cam 02136... 617 864-1535
Roger 150 St Randolph Box 02115... 617 424-6148
Roy «Cassard Ave Cam 02138... 617 491-6115
Royce 18 Salisbury Dr 02129... 617 241-9418

Our Idea: Meaning Through Geography

wide at **SuperPages.com**

195

Car

C

576-1562	Carthage New England Inc.	978 356-9960
30 Allen St Ipswich 01903		
Carthage		
881 447-4101	30 Jewett Rte 02331	978 323-7639
250 259-9961	Carthage Haven	
	9 Beechwood Hwy 01919	617 442-9780
	B H 02326	617 341-5152
576 566-1282	Jessie Jo Stoutcher Chs 02126	617 261-0253
376 364-1338	Lucella 1400 Green Chs 02339	617 576-1061
	M W 0509 Rte 02339	617 323-9713
361-0380	Melvin 151 Green Chs 02339	617 576-1061
576 566-4585	38 Appleton Stn 02134	617 695-6996
628 628-4285	Carthage	617 338-3219
	Cartho. Thos J & R 02134	
	1 Paradise Rd 02134	617 698-6163
576 445-5116	Chatham	
	50 Thompson Ln Mt 02386	617 696-6919
822 822-2979	Chatham	617 327-2257
576 567-9698	Chatham	617 445-5230
	A St Bede's Ws Rduary 02319	617 442-1219
	A 280 Parker Av Cambridge 02142	617 497-1174
676 667-5190	Chatham	617 564-7710
576 569-1471	Adams 351 Centre St Mt 02386	617 698-0974
	Alice 128 Elmwood Stn 02253	617 425-0193
	Chatham	617 425-0193
338 338-9110	Andrew F 82 River Stn Mt 02143	617 623-6971
825 825-9199	Cartho Anne Mt	
	Cartho Atkins	
296 296-1593	Cartho Atkins	617 739-1022
	Cartho Atkins	
676 670-2078	B E of Goodwin Av Mt 02386	617 536-4239
623 623-9001	Cartho Barbara L Mt	617 296-6911
	Cartho Barbara L Mt	
296 296-4275	Cartho	617 636-0051
576 542-1521	Cartho	617 253-4368
	Cartho	
664 664-5232	121 Gladstone Etn 02138	617 567-3040
541 541-5669	Bethesda 25 Melrose Dr 02124	617 567-3040
	Bethesda	617 993-1931
376 376-2662	Carth Broadcasting Co	
	Carth Broadcasting Co	
	58 Park Pl Weymouth 02190	617 423-0210
879 879-0300	Chatham Consultants Inc	
	20 East 2nd Fwyd Av 02138	617 225-1000
576 576-1513	Chatham	617 782-2130
576 569-4119	Chatham	617 669-1545
	Chatham	617 491-8622
509 509-8782	Chatham	617 524-5558
	Chatham	617 524-5558

[illegible][illegible][illegible]

Our Idea: Meaning Through Geography

wide at **SuperPages.com**

195

Car

C

56-156-1282	Cartage New England Inc. 26 Allen St. (see B1833...)	978 356-9960	Carte & Rye 27 Col. Ave. (see B1833...)	617 327-1005	Cartier Nello E. 1332 Washington Ave. (see B1133...)	617 267-6483
56-156-1283	Cartersville 33 Jewett St. (see B1133...)	617 323-7339	Carte's 1000 Washington St. (see B1133...)	617 437-7331	Cartier's 1151 Parkside Ave. (see B1133...)	617 699-3037
56-257-9961	Cartersville 9 Forest St. (see B1133...)	617 442-9780	Francis S. 134 Temple Mt. (see B1133...)	617 323-6781	Nick 21 Parkside Ave. (see B1133...)	617 267-5222
56-156-1282	B. H. (see B1133...)	617 361-5253	Franklin & Ames 27 Mt. Vernon St. (see B1133...)	617 354-0798	Norman G. 160 Herick Rd. Newton (see B1133...)	617 527-0486
56-156-1283	Jessie B. 50 Decatur St. (see B1133...)	617 241-0153	Fred 42 Newbury St. (see B1133...)	617 524-3078	Norfolk 1000 Washington St. (see B1133...)	617 698-0106
56-156-1283	McCalla 134 Temple Mt. (see B1133...)	617 361-5253	Frederic 96 Newbury St. (see B1133...)	617 698-1343	P. A. 1000 Washington St. (see B1133...)	617 698-1343
361-0380	McN 90 Howe St. (see B1133...)	617 323-9713	G. T. 77 Franklin St. (see B1133...)	617 623-7121	P. E. 1000 Washington St. (see B1133...)	617 698-1343
56-156-1283	Melvin 515 Green Comm. (see B1133...)	617 576-1061	Gayle 25 Franklin St. (see B1133...)	617 522-3215	P. E. 1000 Washington St. (see B1133...)	617 698-1343
56-156-1283	84 Apollon Boston (see B1133...)	617 695-6992	George 55 Newbury St. (see B1133...)	617 367-5458	P. L. 440 Franklin St. (see B1133...)	617 327-0176
56-156-1283	Cartersville Cartersville 25 & 4th 17 Franklin St. (see B1133...)	617 338-8219	Cartier Halliday 25 Mount Vernon St. (see B1133...)	617 456-1689	Paul & Constance 1400 Washington St. (see B1133...)	617 327-0176
56-445-5116	Thomas & Cartier 26 Thompson St. (see B1133...)	617 696-6919	Cartier Hardy C. 1400 Washington St. (see B1133...)	617 325-5465	Paul & Constance 1400 Washington St. (see B1133...)	617 327-0176
822-892-2982	Cartier A. (see B1133...)	617 327-2257	Cartier-Hill 144 Summer St. (see B1133...)	617 542-7967	Paul & Constance 1400 Washington St. (see B1133...)	617 327-0176
56-269-768	A. 21 Beaufort Way (see B1133...)	617 442-1219	Horace 43 Newbury St. (see B1133...)	617 576-7520	Cartier Pilgrimage Inc. 11 Beaver G. 731 Franklin St. (see B1133...)	781 235-4488
56-657-519	A. 21 Beaufort Way (see B1133...)	617 442-1219	Howard 43 Newbury St. (see B1133...)	617 542-5307	Cartier Printing 11 Beaver G. 731 Franklin St. (see B1133...)	781 235-4488
56-657-519	A. 21 Beaufort Way (see B1133...)	617 442-1219	J. C. 43 Newbury St. (see B1133...)	617 542-5307	Cartier Printing 11 Beaver G. 731 Franklin St. (see B1133...)	781 235-4488
56-657-519	A. 21 Beaufort Way (see B1133...)	617 442-1219	J. C. 43 Newbury St. (see B1133...)	617 542-5307	Cartier Printing 11 Beaver G. 731 Franklin St. (see B1133...)	781 235-4488
56-657-519	A. 21 Beaufort Way (see B1133...)	617 442-1219	J. C. 43 Newbury St. (see B1133...)	617 542-5307	Cartier Printing 11 Beaver G. 731 Franklin St. (see B1133...)	781 235-4488
56-657-519	A. 21 Beaufort Way (see B1133...)	617 442-1219	J. C. 43 Newbury St. (see B1133...)	617 542-5307	Cartier Printing 11 Beaver G. 731 Franklin St. (see B1133...)	781 235-4488
56-657-519	A. 21 Beaufort Way (see B1133...)	617 442-1219	J. C. 43 Newbury St. (see B1133...)	617 542-5307	Cartier Printing 11 Beaver G. 731 Franklin St. (see B1133...)	781 235-4488
56-657-519	A. 21 Beaufort Way (see B1133...)	617 442-1219	J. C. 43 Newbury St. (see B1133...)	617 542-5307	Cartier Printing 11 Beaver G. 731 Franklin St. (see B1133...)	781 235-4488
56-657-519	A. 21 Beaufort Way (see B1133...)	617 442-1219	J. C. 43 Newbury St. (see B1133...)	617 542-5307	Cartier Printing 11 Beaver G. 731 Franklin St. (see B1133...)	781 235-4488
56-657-519	A. 21 Beaufort Way (see B1133...)	617 442-1219	J. C. 43 Newbury St. (see B1133...)	617 542-5307	Cartier Printing 11 Beaver G. 731 Franklin St. (see B1133...)	781 235-4488
56-657-519	A. 21 Beaufort Way (see B1133...)	617 442-1219	J. C. 43 Newbury St. (see B1133...)	617 542-5307	Cartier Printing 11 Beaver G. 731 Franklin St. (see B1133...)	781 235-4488
56-657-519	A. 21 Beaufort Way (see B1133...)	617 442-1219	J. C. 43 Newbury St. (see B1133...)	617 542-5307	Cartier Printing 11 Beaver G. 731 Franklin St. (see B1133...)	781 235-4488
56-657-519	A. 21 Beaufort Way (see B1133...)	617 442-1219	J. C. 43 Newbury St. (see B1133...)	617 542-5307	Cartier Printing 11 Beaver G. 731 Franklin St. (see B1133...)	781 235-4488
56-657-519	A. 21 Beaufort Way (see B1133...)	617 442-1219	J. C. 43 Newbury St. (see B1133...)	617 542-5307	Cartier Printing 11 Beaver G. 731 Franklin St. (see B1133...)	781 235-4488
56-657-519	A. 21 Beaufort Way (see B1133...)	617 442-1219	J. C. 43 Newbury St. (see B1133...)	617 542-5307	Cartier Printing 11 Beaver G. 731 Franklin St. (see B1133...)	781 235-4488
56-657-519	A. 21 Beaufort Way (see B1133...)	617 442-1219	J. C. 43 Newbury St. (see B1133...)	617 542-5307	Cartier Printing 11 Beaver G. 731 Franklin St. (see B1133...)	781 23



→ We develop a (conceptual) geography of clusterings

A New Strategy

A New Strategy

- 1 **Code text as numbers** (in one *or more* of several ways)

A New Strategy

- ① **Code text as numbers** (in one *or more* of several ways)
- ② **Apply all existing clustering methods** (that have been used by at least one person other than the author) to the data — each representing different substantive assumptions (<15 mins)

A New Strategy

- 1 Code text as numbers (in one *or more* of several ways)
- 2 Apply all existing clustering methods (that have been used by at least one person other than the author) to the data — each representing different substantive assumptions (<15 mins)
- 3 Develop an application-independent distance metric between clusterings

A New Strategy

- ① **Code text as numbers** (in one *or more* of several ways)
- ② **Apply all existing clustering methods** (that have been used by at least one person other than the author) to the data — each representing different substantive assumptions (<15 mins)
- ③ Develop an **application-independent distance** metric between clusterings
- ④ Create a **metric space of clusterings**, and a 2D projection

A New Strategy

- 1 Code text as numbers (in one or more of several ways)
- 2 Apply all existing clustering methods (that have been used by at least one person other than the author) to the data — each representing different substantive assumptions (<15 mins)
- 3 Develop an application-independent distance metric between clusterings
- 4 Create a metric space of clusterings, and a 2D projection
- 5 Introduce the local cluster ensemble to summarize any point, including points with no existing clustering

A New Strategy

- 1 **Code text as numbers** (in one *or more* of several ways)
- 2 **Apply all existing clustering methods** (that have been used by at least one person other than the author) to the data — each representing different substantive assumptions (<15 mins)
- 3 Develop an **application-independent distance** metric between clusterings
- 4 Create a **metric space of clusterings**, and a 2D projection
- 5 Introduce the **local cluster ensemble** to summarize any point, including points with no existing clustering
- 6 Propose a new **animated visualization**: use the local cluster ensemble to explore the space of clusterings (smoothly morphing from one into others)

A New Strategy

- 1 Code text as numbers (in one or more of several ways)
- 2 Apply all existing clustering methods (that have been used by at least one person other than the author) to the data — each representing different substantive assumptions (<15 mins)
- 3 Develop an application-independent distance metric between clusterings
- 4 Create a metric space of clusterings, and a 2D projection
- 5 Introduce the local cluster ensemble to summarize any point, including points with no existing clustering
- 6 Propose a new animated visualization: use the local cluster ensemble to explore the space of clusterings (smoothly morphing from one into others)

⇒ meaning revealed through a geography of clusterings

You choose one (or more), based on insight, discovery, useful information,...



Application-Independent Distance Metric: Axioms

Application-Independent Distance Metric: Axioms

- 1 Clusterings with more **pairwise document agreements** are closer (we prove: pairwise agreements encompass triples, quadruples, etc.)

Application-Independent Distance Metric: Axioms

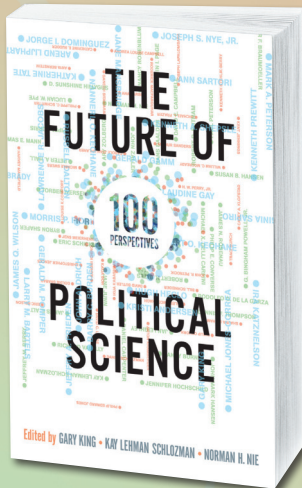
- 1 Clusterings with more **pairwise document agreements** are closer (we prove: pairwise agreements encompass triples, quadruples, etc.)
- 2 **Invariance**: Distance is invariant to the number of documents (for any fixed number of clusters)

Application-Independent Distance Metric: Axioms

- ① Clusterings with more **pairwise document agreements** are closer (we prove: pairwise agreements encompass triples, quadruples, etc.)
- ② **Invariance**: Distance is invariant to the number of documents (for any fixed number of clusters)
- ③ **Scale**: the maximum distance is set to $\log(\text{num clusters})$

Application-Independent Distance Metric: Axioms

- ① Clusterings with more **pairwise document agreements** are closer (we prove: pairwise agreements encompass triples, quadruples, etc.)
 - ② **Invariance**: Distance is invariant to the number of documents (for any fixed number of clusters)
 - ③ **Scale**: the maximum distance is set to $\log(\text{num clusters})$
- ↪ **Only one measure satisfies all three** (the “variation of information”)



Available March 2009: 304pp
 Pb: 978-0-415-99701-0: **\$24.95**
www.routledge.com/politics

THE FUTURE OF POLITICAL SCIENCE

100 Perspectives

Edited by Gary King, Harvard University, Kay Lehman Schlozman, Boston College
 and Norman H. Nie, Stanford University

"The list of authors in *The Future of Political Science* is a 'who's who' of political science. As I was reading it, I came to think of it as a platter of tasty hors d'oeuvres. It hooked me thoroughly."

—Peter Kingstone, University of Connecticut

"In this one-of-a-kind collection, an eclectic set of contributors offer short but forceful forecasts about the future of the discipline. The resulting assortment is captivating, consistently thought-provoking, often intriguing, and sure to spur discussion and debate."

—Wendy K. Tam Cho, University of Illinois at Urbana-Champaign

"King, Schlozman, and Nie have created a visionary and stimulating volume. The organization of the essays strikes me as nothing less than brilliant. . . It is truly a joy to read."

—Lawrence C. Dodd, Manning J. Dauer Eminent Scholar in Political Science,
 University of Florida

Routledge
 Taylor & Francis Group
 an informa business

Evaluators' Rate Machine Choices Better Than Their Own

Evaluators' Rate Machine Choices Better Than Their Own

- Scale: (1) unrelated, (2) loosely related, or (3) closely related

Evaluators' Rate Machine Choices Better Than Their Own

- Scale: (1) unrelated, (2) loosely related, or (3) closely related
- Table reports: mean(scale)

Evaluators' Rate Machine Choices Better Than Their Own

- Scale: (1) unrelated, (2) loosely related, or (3) closely related
- Table reports: mean(scale)

Pairs from	Overall Mean	Evaluator 1	Evaluator 2
Random Selection	1.38	1.16	1.60

Evaluators' Rate Machine Choices Better Than Their Own

- Scale: (1) unrelated, (2) loosely related, or (3) closely related
- Table reports: mean(scale)

Pairs from	Overall Mean	Evaluator 1	Evaluator 2
Random Selection	1.38	1.16	1.60

Evaluators' Rate Machine Choices Better Than Their Own

- Scale: (1) unrelated, (2) loosely related, or (3) closely related
- Table reports: mean(scale)

Pairs from	Overall Mean	Evaluator 1	Evaluator 2
Random Selection	1.38	1.16	1.60
Hand-Coded Clusters	1.58	1.48	1.68

Evaluators' Rate Machine Choices Better Than Their Own

- Scale: (1) unrelated, (2) loosely related, or (3) closely related
- Table reports: mean(scale)

Pairs from	Overall Mean	Evaluator 1	Evaluator 2
Random Selection	1.38	1.16	1.60
Hand-Coded Clusters	1.58	1.48	1.68
Hand-Coding	2.06	1.88	2.24

Evaluators' Rate Machine Choices Better Than Their Own

- Scale: (1) unrelated, (2) loosely related, or (3) closely related
- Table reports: mean(scale)

Pairs from	Overall Mean	Evaluator 1	Evaluator 2
Random Selection	1.38	1.16	1.60
Hand-Coded Clusters	1.58	1.48	1.68
Hand-Coding	2.06	1.88	2.24
Machine	2.24	2.08	2.40

Evaluators' Rate Machine Choices Better Than Their Own

- Scale: (1) unrelated, (2) loosely related, or (3) closely related
- Table reports: mean(scale)

Pairs from	Overall Mean	Evaluator 1	Evaluator 2
Random Selection	1.38	1.16	1.60
Hand-Coded Clusters	1.58	1.48	1.68
Hand-Coding	2.06	1.88	2.24
Machine	2.24	2.08	2.40

p.s. The hand-coders did the evaluation!

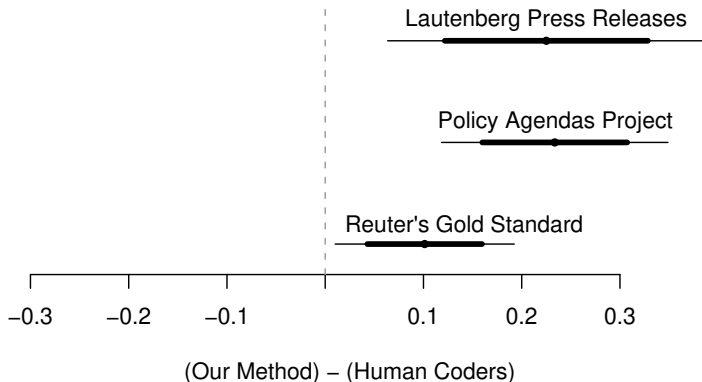
Cluster Quality Experiments

Cluster Quality Experiments

Scale: $\text{mean}(\text{within clusters}) - \text{mean}(\text{between clusters})$

Cluster Quality Experiments

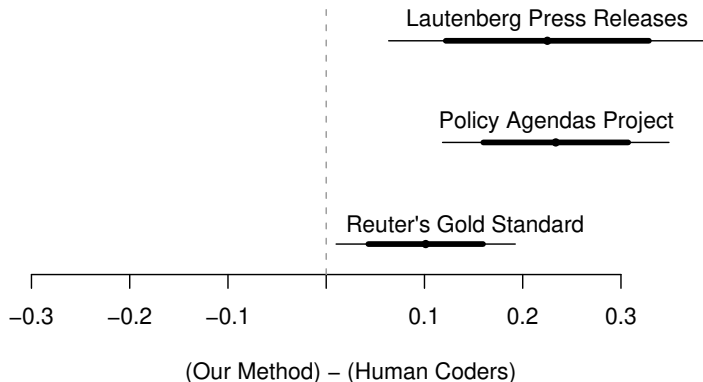
Scale: $\text{mean}(\text{within clusters}) - \text{mean}(\text{between clusters})$



Lautenberg: 200 Senate Press Releases (appropriations, economy, education, tax, veterans, ...)

Cluster Quality Experiments

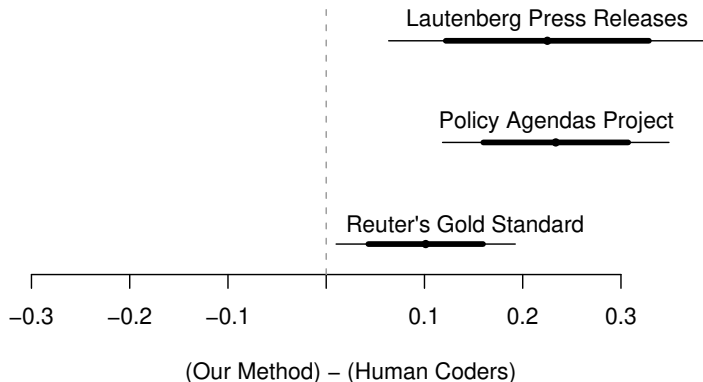
Scale: $\text{mean}(\text{within clusters}) - \text{mean}(\text{between clusters})$



Policy Agendas: 213 quasi-sentences from Bush's State of the Union (agriculture, banking & commerce, civil rights/liberties, defense, ...)

Cluster Quality Experiments

Scale: $\text{mean}(\text{within clusters}) - \text{mean}(\text{between clusters})$



Reuter's: financial news (trade, earnings, copper, gold, coffee, ...); "gold standard" for supervised learning studies

What do Members of Congress Do?

What do Members of Congress Do?

Substantive example of a finding, using our approach

What do Members of Congress Do?

Substantive example of a finding, using our approach

- David Mayhew's (1974) famous typology

What do Members of Congress Do?

Substantive example of a finding, using our approach

- David Mayhew's (1974) famous typology
 - 1 Advertising

What do Members of Congress Do?

Substantive example of a finding, using our approach

- David Mayhew's (1974) famous typology
 - 1 Advertising
 - 2 Credit Claiming

What do Members of Congress Do?

Substantive example of a finding, using our approach

- David Mayhew's (1974) famous typology
 - 1 Advertising
 - 2 Credit Claiming
 - 3 Position Taking

What do Members of Congress Do?

Substantive example of a finding, using our approach

- David Mayhew's (1974) famous typology
 - ① Advertising
 - ② Credit Claiming
 - ③ Position Taking
- We find one more: **Partisan Taunting**

What do Members of Congress Do?

Substantive example of a finding, using our approach

- David Mayhew's (1974) famous typology
 - ① Advertising
 - ② Credit Claiming
 - ③ Position Taking
- We find one more: **Partisan Taunting**
 - "Senator Lautenberg Blasts Republicans as 'Chicken Hawks' "
[Government Oversight]

What do Members of Congress Do?

Substantive example of a finding, using our approach

- David Mayhew's (1974) famous typology
 - ① Advertising
 - ② Credit Claiming
 - ③ Position Taking
- We find one more: **Partisan Taunting**
 - "Senator Lautenberg Blasts Republicans as 'Chicken Hawks' "
[Government Oversight]
 - "The scopes trial took place in 1925. Sadly, President Bush's veto today shows that we haven't progressed much since then." [Healthcare]

What do Members of Congress Do?

Substantive example of a finding, using our approach

- David Mayhew's (1974) famous typology
 - ① Advertising
 - ② Credit Claiming
 - ③ Position Taking
- We find one more: **Partisan Taunting**
 - "Senator Lautenberg Blasts Republicans as 'Chicken Hawks' " [Government Oversight]
 - "The scopes trial took place in 1925. Sadly, President Bush's veto today shows that we haven't progressed much since then." [Healthcare]
 - "John Kerry had enough conviction to sign up for the military during wartime, unlike the Vice President, who had a deep conviction to avoid military service" [Government Oversight]

What do Members of Congress Do?

Substantive example of a finding, using our approach

- David Mayhew's (1974) famous typology
 - ① Advertising
 - ② Credit Claiming
 - ③ Position Taking
- We find one more: **Partisan Taunting**
 - "Senator Lautenberg Blasts Republicans as 'Chicken Hawks' " [Government Oversight]
 - "The scopes trial took place in 1925. Sadly, President Bush's veto today shows that we haven't progressed much since then." [Healthcare]
 - "John Kerry had enough conviction to sign up for the military during wartime, unlike the Vice President, who had a deep conviction to avoid military service" [Government Oversight]
 - ~> Is this what "party in government" means?

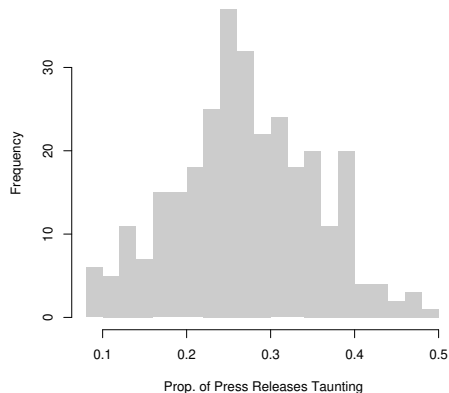
Partisan Taunting Hypothesis Verification

Partisan Taunting Hypothesis Verification

Discovered in $n = 200$ (1 senator); confirmed in $n = 64,033$ (301 senator-years)

Partisan Taunting Hypothesis Verification

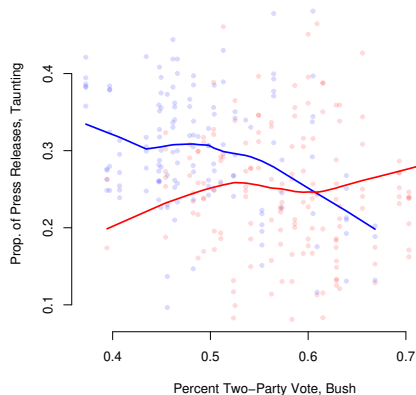
Discovered in $n = 200$ (1 senator); confirmed in $n = 64,033$ (301 senator-years)



Prevalence of partisan taunting (it happens a lot!)

Partisan Taunting Hypothesis Verification

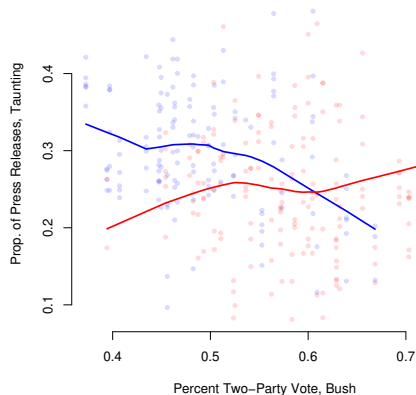
Discovered in $n = 200$ (1 senator); confirmed in $n = 64,033$ (301 senator-years)



Partisan taunting occurs more in lopsided districts

Partisan Taunting Hypothesis Verification

Discovered in $n = 200$ (1 senator); confirmed in $n = 64,033$ (301 senator-years)



↪ Conflict: deliberation (no taunting) v. reflection (lopsided districts)

More Informative Discoveries

More Informative Discoveries

- Found 2 scholars analyzing lots of textual data for their work

More Informative Discoveries

- Found 2 scholars analyzing lots of textual data for their work
- For each: created 2 clusterings from each of 3 methods, including ours

More Informative Discoveries

- Found 2 scholars analyzing lots of textual data for their work
- For each: created 2 clusterings from each of 3 methods, including ours
- Created info packet on each clustering (for each cluster: exemplar document, automated content summary)

More Informative Discoveries

- Found 2 scholars analyzing lots of textual data for their work
- For each: created 2 clusterings from each of 3 methods, including ours
- Created info packet on each clustering (for each cluster: exemplar document, automated content summary)
- Asked for $\binom{6}{2}=15$ pairwise comparisons

More Informative Discoveries

- Found 2 scholars analyzing lots of textual data for their work
- For each: created 2 clusterings from each of 3 methods, including ours
- Created info packet on each clustering (for each cluster: exemplar document, automated content summary)
- Asked for $\binom{6}{2}=15$ pairwise comparisons
- Both cases a Condorcet winner:

More Informative Discoveries

- Found 2 scholars analyzing lots of textual data for their work
- For each: created 2 clusterings from each of 3 methods, including ours
- Created info packet on each clustering (for each cluster: exemplar document, automated content summary)
- Asked for $\binom{6}{2}=15$ pairwise comparisons
- Both cases a Condorcet winner:

“Immigration”:

Our Method 1 → vMF 1 → vMF 2 → Our Method 2 → K-Means 1 → K-Means 2

More Informative Discoveries

- Found 2 scholars analyzing lots of textual data for their work
- For each: created 2 clusterings from each of 3 methods, including ours
- Created info packet on each clustering (for each cluster: exemplar document, automated content summary)
- Asked for $\binom{6}{2}=15$ pairwise comparisons
- Both cases a Condorcet winner:

“Immigration”:

Our Method 1 → vMF 1 → vMF 2 → Our Method 2 → K-Means 1 → K-Means 2

“Genetic testing”:

Our Method 1 → {Our Method 2, K-Means 1, K-means 2} → Dir Proc. 1 → Dir Proc. 2

Last points

Last points

- Intended contributions:

Last points

- **Intended contributions:**
 - An encompassing cluster analytic approach for discovery

Last points

- **Intended contributions:**
 - An encompassing cluster analytic approach for discovery
 - A new approach to evaluating results in unsupervised learning

- **Intended contributions:**
 - An encompassing cluster analytic approach for discovery
 - A new approach to evaluating results in unsupervised learning
 - Especially useful for the ongoing spectacular increase in the production and availability of unstructured text

- **Intended contributions:**
 - An encompassing cluster analytic approach for discovery
 - A new approach to evaluating results in unsupervised learning
 - Especially useful for the ongoing spectacular increase in the production and availability of unstructured text
- **Future research:**

- **Intended contributions:**

- An encompassing cluster analytic approach for discovery
- A new approach to evaluating results in unsupervised learning
- Especially useful for the ongoing spectacular increase in the production and availability of unstructured text

- **Future research:**

- Advancing our approach:

- **Intended contributions:**

- An encompassing cluster analytic approach for discovery
- A new approach to evaluating results in unsupervised learning
- Especially useful for the ongoing spectacular increase in the production and availability of unstructured text

- **Future research:**

- Advancing our approach: (1) $>2D$ exploration,

- **Intended contributions:**

- An encompassing cluster analytic approach for discovery
- A new approach to evaluating results in unsupervised learning
- Especially useful for the ongoing spectacular increase in the production and availability of unstructured text

- **Future research:**

- Advancing our approach: (1) $>2D$ exploration, (2) alternative visualizations of the space of clusterings,

- **Intended contributions:**

- An encompassing cluster analytic approach for discovery
- A new approach to evaluating results in unsupervised learning
- Especially useful for the ongoing spectacular increase in the production and availability of unstructured text

- **Future research:**

- Advancing our approach: (1) $>2D$ exploration, (2) alternative visualizations of the space of clusterings, (3) including more methods

- **Intended contributions:**

- An encompassing cluster analytic approach for discovery
- A new approach to evaluating results in unsupervised learning
- Especially useful for the ongoing spectacular increase in the production and availability of unstructured text

- **Future research:**

- Advancing our approach: (1) $>2D$ exploration, (2) alternative visualizations of the space of clusterings, (3) including more methods
- Evaluating new individual methods:

- **Intended contributions:**

- An encompassing cluster analytic approach for discovery
- A new approach to evaluating results in unsupervised learning
- Especially useful for the ongoing spectacular increase in the production and availability of unstructured text

- **Future research:**

- Advancing our approach: (1) $>2D$ exploration, (2) alternative visualizations of the space of clusterings, (3) including more methods
- Evaluating new individual methods: (1) distance from existing methods and their averages,

- **Intended contributions:**

- An encompassing cluster analytic approach for discovery
- A new approach to evaluating results in unsupervised learning
- Especially useful for the ongoing spectacular increase in the production and availability of unstructured text

- **Future research:**

- Advancing our approach: (1) $>2D$ exploration, (2) alternative visualizations of the space of clusterings, (3) including more methods
- Evaluating new individual methods: (1) distance from existing methods and their averages, (2) usefulness of discoveries in given data sets.

For more information:

<http://GKing.Harvard.edu>